Ahmedabad University

Milestone 2

Cross-Environment and SNR Robustness in Wi-Fi CSI-Based Human Activity Recognition: A Benchmark Study and Lightweight Domain-Shift Analysis

Category

AI Algorithms + Existing Dataset

Group 7

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github.com/Pranavv26/wifi-csi-domain-robustness

Abstract

Models for Wi-Fi channel state information (CSI) human activity recognition (HAR) degrade when deployed in unseen environments or on unseen users. Our Milestone 1 (M1) report [2] scoped this project as a reproduction-oriented study of DATTA [1] on Widar3.0-G6D. This Milestone 2 (M2) report covers only the first stage: a technically sound cross-domain pipeline and the vanilla WiFlexFormer baseline (W). We describe the preprocessing, the source/held-out evaluation protocol, and the baseline training procedure. Our team obtained a best source-validation macro-F1 of 48.81% (epoch 5) and a final held-out macro-F1 of 45.96% (accuracy 43.34%). We compare these figures with the published W results in [1] and explain why the comparison is not strictly like-for-like. No DATTA component beyond W has been implemented at this stage.

1. Objective and Continuity from M1

M1 [2] defined the problem, motivated DATTA [1] as the most relevant recent reference for cross-domain Wi-Fi HAR, and planned a staged reproduction. DATTA combines domain-adversarial training (DAT), test-time adaptation (TTA), and random weight resetting on a lightweight WiFlexFormer backbone [1], [4]. M2 follows the plan in [2]: before making any claim about these components, we establish that the data pipeline, split discipline, and baseline evaluation are sound.

M2 is limited to the vanilla baseline W. CSI augmentation, DAT with the gradient reversal layer (GRL), the confidence-control constraint, TTA, and random weight resetting have **not** been implemented; they are scheduled for Milestone 3 (M3). Our contribution is a reproduction-oriented baseline experiment, not a new method.

2. Experimental Pipeline

A. Dataset preparation

We use Widar3.0-G6D, the six-gesture subset of Widar3.0 [3] prepared for cross-domain evaluation in [1]. The six classes are push and pull, sweep, clap, slide, draw circle, and draw zigzag. Preprocessing follows the DATTA preparation protocol [1]:

- **Input:** CSI amplitude from 30 subcarriers of the first antenna.
- **Temporal subsampling:** to 100 Hz.
- **Sequence length:** sequences of 120–220 packets (about 1.2–2.2 s) are retained, and shorter sequences are zero-padded.
- **Normalization:** min-max scaling of the amplitude features.

A domain in this benchmark is a unique room–participant combination. Source and held-out

subsets are disjoint by domain, so the held-out evaluation measures generalization to unseen domains rather than within-domain memorization [1].

B. Evaluation protocol

Table 1: Dataset split and evaluation protocol.

Subset	Samples	Role
Source training	19,585	Fit model weights
Source validation	4,897	Checkpoint selection (macro-F1 only)
Held-out unseen-domain	34,166	Final evaluation, performed once

Model selection uses validation macro-F1 only, and the held-out set plays no role in selecting the checkpoint or any training choice. After the checkpoint was fixed, the held-out set was evaluated once.

We prioritize macro-F1 because it averages per-class F1 with equal weight, so a model cannot score well by performing well on a few classes while failing on others. Accuracy can hide such behavior. Under domain shift, per-class degradation is common, and the DATTA authors motivate their confidence-control constraint by the problem of over-confident, class-imbalanced predictions [1].

3. Baseline Configuration

W is the vanilla WiFlexFormer: a convolutional stem, Gaussian positional encoding, and a transformer encoder with a class token and a linear classification layer [1], [4]. It runs in amplitude mode, without augmentation, domain labels, an adversarial branch, or test-time adaptation.

We trained W for 15 epochs on CPU with PyTorch 2.14.1+cpu. The configuration was:

- **Optimizer:** AdamW, initial learning rate 0.001, weight decay 0.0001.
- **Batch size and seed:** batch size 8, random seed 42.
- **Learning-rate schedule:** ReduceLROnPlateau monitoring validation macro-F1, halving the learning rate after two epochs without improvement.
- **Validation split:** 20% of the source-domain training cache was reserved for validation.
- **Checkpoint selection:** the checkpoint with the highest validation macro-F1 was saved and used for the single final held-out evaluation.

The scheduler uses only source-validation information, so the held-out data remain untouched during training. We make no claim that these settings match the authors' hyperparameters, which [1] lists in its supplementary material.

4. Results

Table 2: Team-obtained (reproduced) and published baseline results. Published values are from [1] (mean ± std over three runs); team values come from a single run (seed 42), with the held-out set evaluated once on the selected checkpoint.

Source / Configuration	Accuracy (%)	Macro-F1 (%)	Interpretation
Team (obtained): W, source validation, epoch 5	44.50	48.81	Highest validation macro-F1; used for checkpoint selection
Team (obtained): W, held-out unseen domains (34,166 samples)	43.34	45.96	Final result; 2.85 pp below the selected validation score
Published [1]: W, no augmentation	38.69 ± 3.17	40.62 ± 2.93	Reference point; different evaluation set and run protocol
Published [1]: W, with augmentation	47.75 ± 4.31	49.32 ± 4.45	Reference point; uses augmentation, which our W does not

Note: [1] labels this metric "F1-Score" and does not state the averaging scheme in the text we reviewed; we treat it as the nearest comparable quantity.

Key observations from our run:

- The best validation macro-F1 (48.81%) occurred at epoch 5, with validation accuracy of 44.50% at that epoch.
- Held-out accuracy was 43.34% and macro-F1 was 45.96%, so macro-F1 decreased by 2.85 percentage points from the selected validation checkpoint to the held-out evaluation.
- Training loss declined from 1.3173 (epoch 1) to 1.2165 (epoch 15).
- Validation macro-F1 fluctuated across epochs but stayed near 48% in later epochs.

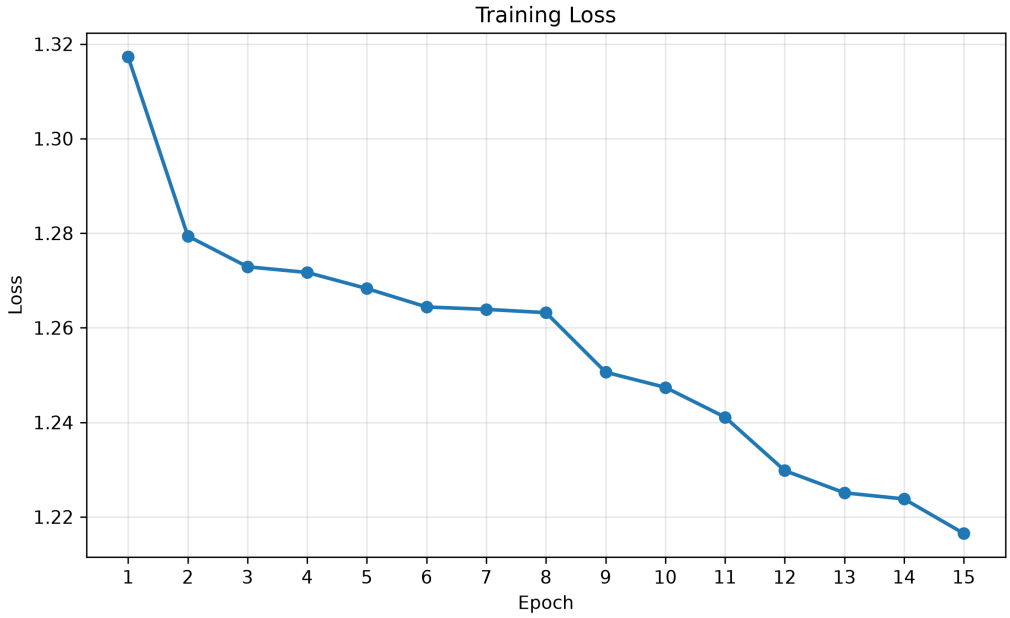


Figure 1: Training loss versus epoch for the vanilla WiFlexFormer baseline (15 epochs).

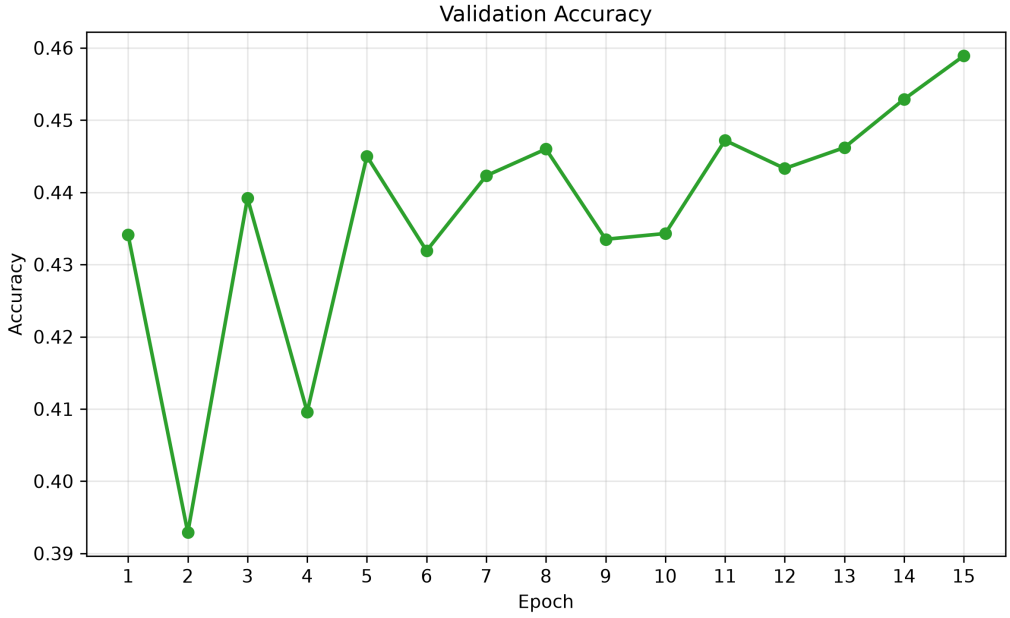


Figure 2: Validation accuracy versus epoch. Accuracy at the selected epoch 5 is 44.50%.

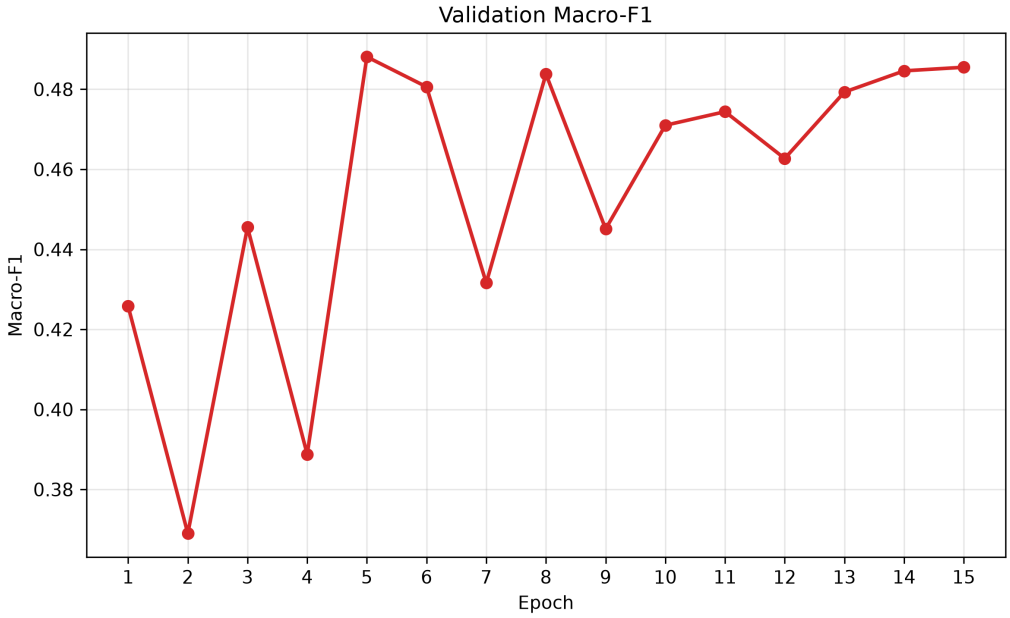


Figure 3: Validation macro-F1 versus epoch. Maximum 48.81% at epoch 5, used for checkpoint selection.

5. Discussion and Reproducibility Notes

Learning is modest. The training loss decreased by about 0.10 over 15 epochs and remained high at the end. The validation macro-F1 curve is somewhat variable and stays near 48% after its epoch-5 peak. We do not interpret this as convergence to an optimum, and we have not diagnosed whether other settings would change it.

The 2.85-point gap should be read cautiously. The validation set is drawn from the source cache, and its score is already low, so a small drop to the held-out set does not by itself indicate robustness. The validation score is also the maximum over 15 epochs, and validation macro-F1 additionally drives the learning-rate scheduler, so it is mildly optimistic by construction. We report the gap as a descriptive measurement, not as evidence of good generalization.

Reproducibility notes:

- Our training and validation counts (19,585 and 4,897) differ from the published counts (19,586 and 4,896) [1] by one sample each, with the same combined total of 24,482. A 20% hold-out of 24,482 samples is 4,896.4, so the difference is consistent with rounding in the validation split. We did not verify this against the authors' partition.
- Selection used validation macro-F1 only, and the held-out set was evaluated once.
- All results come from one run (seed 42) on CPU, so run-to-run variance is not quantified.
- We did not measure per-class metrics, parameter count, or inference time in M2, and we do not report a confusion matrix. These are left to later milestones.

6. SOTA Positioning

DATTA [1] remains the most directly relevant recent benchmark for real-time cross-domain Wi-Fi HAR on Widar3.0-G6D. The authors report that the full method reaches 68.13% F1, compared with 40.62% for W without augmentation and 49.32% for W with augmentation (three-run averages on their shuffled-Test protocol) [1]. These are **published** results, not ours.

Our **obtained** 45.96% macro-F1 lies between the two published W reference points. It is not a strict apples-to-apples comparison, for three reasons:

1. Our held-out set (34,166 samples) equals the paper's separate Val_TTA (3,417) and Test (30,749) portions combined [1].
2. The published figures are three-run means with standard deviations of 3–4 points, whereas ours come from a single run.
3. Our W uses no augmentation, whereas one of the two published reference points uses it.

We therefore make no claim that our baseline matches, surpasses, or disproves DATTA. In M3 we will preserve Val_TTA as a separate subset from the final Test set.

7. Conclusion and Next Steps

M2 establishes the preprocessing pipeline, the vanilla W baseline, the evaluation procedure, a saved checkpoint, the metrics, and the plots needed for controlled M3 work. Next steps are:

1. Separate Val_TTA from the final Test set.
2. Add CSI augmentation.
3. Implement domain-adversarial training.
4. Add TTA and random weight resetting, tuning only on Val_TTA.

Limitations. These results come from one baseline configuration (seed 42) trained on CPU for 15 epochs, so run-to-run variance is not quantified. The held-out set merges what [1] treats as two separate subsets, which limits comparability with the published numbers. We have no per-class analysis, parameter count, or runtime measurement. The baseline's modest performance means that any claim about DATTA components must wait for the controlled M3 experiments.

References

[1] J. Strohmayer, R. Sterzinger, M. Wödlinger, and M. Kampel, "DATTA: Domain-Adversarial Test-Time Adaptation for Cross-Domain WiFi-Based Human Activity Recognition," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2026. Code: https://github.com/StrohmayerJ/DATTA

[2] P. Jain, P. Jawaharani, and K. Lunkad, "Milestone 1: Cross-Environment and SNR Robustness in Wi-Fi CSI-Based Human Activity Recognition," ECE 310 Wireless Communications, Ahmedabad University, Group 7, 2026.

[3] Y. Zhang, Y. Zheng, K. Qian, G. Zhang, Y. Liu, C. Wu, and Z. Yang, "Widar3.0: Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8671–8688, 2022.

[4] J. Strohmayer, M. Wödlinger, and M. Kampel, "WiFlexFormer: Efficient WiFi-Based Person-Centric Sensing," 2024.